



The \$state Rune

Q34

SVELTE 5 RUNES STATE MANAGEMENT

INTERVIEW QUESTION

Question. Explain the core purpose, syntax, and execution model of the `$state` rune.

ANSWER. The `$state` rune is the fundamental building block of reactivity in Svelte 5.

1. **Core Purpose:** It declares a variable as reactive. Unlike standard JavaScript variables, changing a `$state` variable guarantees that any part of the UI or derived logic depending on it will automatically update. 2. **Syntax:** It is invoked as a function wrapping the initial value: `let count = $state(0);`. 3. **Execution Model:** Under the hood, `$state` creates a reactive "signal." When the variable is read during a render, Svelte records the DOM node as a dependency. When the variable is updated, `$state` acts like a broadcaster, sending a direct, fine-grained signal to update *only* those specific DOM nodes, bypassing full component re-renders entirely.

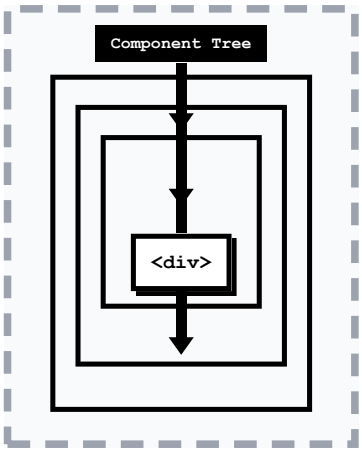
Broadcaster.svelte

```
<script>
  // Declaring a live signal
  let message = $state('Hello!');

  function update() {
    // Re-assigning triggers the broadcast
    message = 'Runes are active.';
  }
</script>

<!-- This specific node listens to the 'message' signal -->
<h1>{message}</h1>
<button onclick={update}>Update</button>
```

LEGACY: TOP-DOWN



RUNES: DIRECT UPDATE

